LAB Assignment 3

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Swayamsiddha Company Data Mart Architecture

a) Identify Grains

1) What is the average time (lead time) between placing a raw material purchase order and the actual delivery date?

2) Which supplier currently has the highest defect or rejection rate during raw material quality inspection?

3) How has the fluctuating cost of raw packaging material impacted overall procurement spending this quarter?

4) What is average transit time of internal stock transfer between the manufacting plant and the regional distribution centers?

5) What is the average Days Sales outstanding (DSO) for our wholesale retail partners?

6) Which specific facility has the highest operational overhead costs per unit manufactered?

B) Identify Dimensions and its type

a) Product :
Type : Slow changing Dimension
attributes : product_id (Primary key) , Product_Name , Category
Brand , sub-category , Unit-cost , unit_weight
MRP , Expiry_date , stock/Inventory

b) Retailer :
Type : Conformed Dimension
attributes : retailer_id (Primary key) , Retailer_Name , Shop-type ,
Store-size , Contact , Category , Status , location

c) Date :
Type : Role-playing Dimension
attribute : date_key , Querter , Year , month , week , Day_Name , Season
, Season_flag , Festival_Name .

d) Promotion :
Type : Low Cardinality Dimension
attribute : Promotion_id (Primary key) , Promotion_name , Discount_type ,
Discount_value , Minimum_Purchase_Quantity.

c) Identify Measures and facts and its types

Fact Table : Fact_Retail_Sales
Fact Table type : Transactional fact Table
↳ Capture Quantitative measurements resulting from individual sales event.

Measures

① Sales_Quantity :
Type : Additive measures / Integer

② Gross_Sales_Amount :
Types : Additive measures / Decimal

③ Discount_Amount :
Types : Additive measures / Decimal

④ Net_sales
Types : Additive measure / Decimal.

⑤ Unit_Retail_price
Type : Non-additive measure / Decimal.

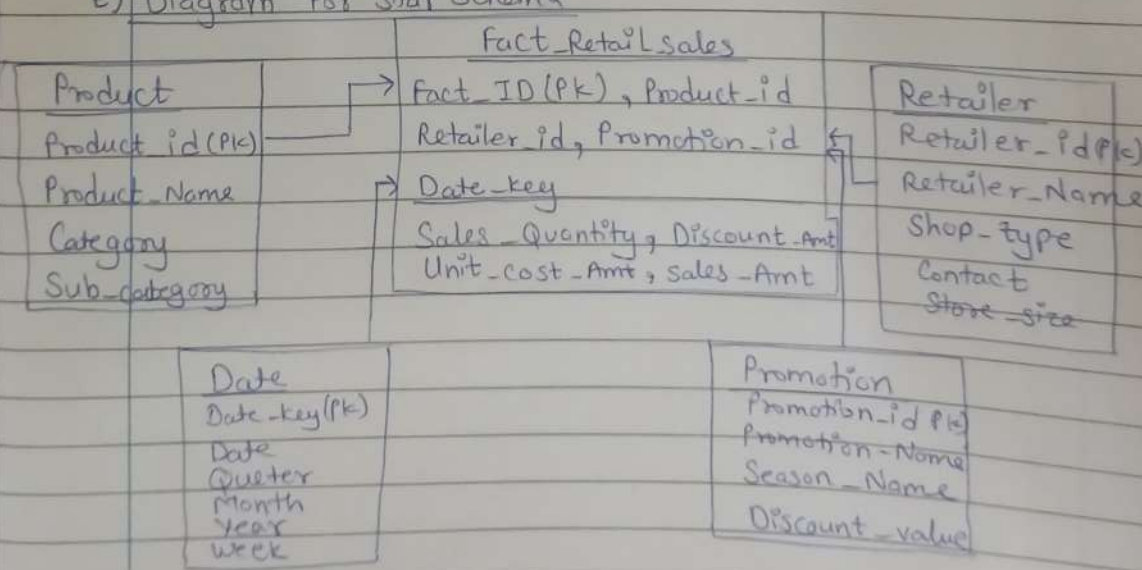
D) Identification of Schema Modeling & Justification.

Model : Star Schema Model

Justification :

- Optimal Query Performance
  In this particular case we need high-speed analytical queries accross sales regions in Maharashtra and product categories. Star schema minimizes table joins significantly impro response time
- Denormalized Dimension of fast Aggregation.
  Dimensions like Product and date are kept denormalized This avoid complex multitable joins required in snow flake Schema
- Intuitive Business Alignment:
  Direct relationship between entities and central sales transaction makes analysis and execution straightforward.

E) Diagram for star Schema



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f) Advantage & Disadvantages of star schema.

Advantage	Disadvantage
→ High Query Response Speed it has minimal joins allowing rapid sales reports	→ Data Redundancy: Denormilization of dimension leads to duplicate text
→ Intuitive Schema structure: orginization can easily understa since table represent real world entities	→ SCD management Overhead: Tracking historical product prices revisions requires implementing Complex SCD logics
→ Direct Compatibility: Highly Conmpatible with mordem BI softoars / tools	→ Update Anomalies.